Calculus II

Problems for Training #5 17–21 Mar 2025

Topics:

- Extreme values and saddle points
- Optimization problems

In Problems 1–3, find all the local maxima, local minima, and saddle points of the functions.

Problem 1. $f(x, y) = 2xy - x^2 - 2y^2 + 3x + 4$.

Problem 2. $f(x, y) = x^2 - y^2 - 2x + 4y + 6$.

Problem 3. $f(x, y) = \frac{1}{x^2+y^2-1}$.

Problem 4. Find the point on the plane $3x + 2y + z = 6$ that is nearest the origin.

Problem 5. Find the minimum distance from the cone $z = \sqrt{x^2 + y^2}$ to the point $(-6, 4, 0)$.

Problem 6. *Extrema on a circle* Find the maximum and minimum values of $x^2 + y^2$ subject to the constraint $x^2 - 2x + y^2 - 4y = 0$.

Problem 7. *Ant on a metal plate* The temperature at a point (x, y) on a metal plate is $T(x, y) = 4x^2 - 4xy + y^2$. An ant on the plate walks around the circle of radius 5 centered at the origin. What are the highest and lowest temperatures encountered by the ant?

Problem 8. *Minimum distance to a point* Find the point on the plane $x+2y+3z = 13$ closest to the point $(1, 1, 1)$.

Problem 9. *Minimum distance to the origin* Find the minimum distance from the surface $x^2 - y^2 - z^2 = 1$ to the origin.

Problem 10. *Minimizing a sum of squares* Find three real numbers whose sum is 9 and the sum of whose squares is as small as possible.

Problem 11. *Rectangular box of largest volume in a sphere* Find the dimensions of the closed rectangular box with maximum volume that can be inscribed in the unit sphere.

END

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Answers to selected problems

1. $(3, \frac{3}{2})$
2. $(1, 2)$
4. $(\frac{9}{7}, \frac{6}{7}, \frac{3}{7})$
5. $\sqrt{26}$
6. $\max = 20$, $\min = 0$
7. $\max = 125$, $\min = 0$
8. $(\frac{3}{2}, 2, \frac{5}{2})$
9. 1
10. $x = y = z = 3$
11. $x = y = z = \dfrac{1}{\sqrt{3}}$.